

Omar Abdul Qadir

Embedded Software Engineer

Contact Info:

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Summary:

Embedded Software Engineer with 3+ years of experience building firmware, RTOS-based systems, and Embedded Linux platforms for smart-grid and communication applications. Focused on low-level driver development, system optimization, and hardware bring-up, with a track record of improving reliability and performance under real-world constraints. Looking to bring that depth to embedded software and firmware roles, while continuing to learn and build reliable, high-quality systems.

Work Experience:

Embedded Software Engineer

Pylon

Nov 2023 – Present

- Developed and optimized communication stacks, including PLC (PRIME I.3.6), DLMS-HDL, and TCP/IP over fiber and copper, reaching 99% data transmission reliability across smart meter networks.
- Designed low-level drivers for UART, SPI, and I2C peripherals, enabling data transfer to EEPROM/Flash memory and NFC/RS485 communication modules.
- Built system-level services for memory management, buffering, and storage integrity, preventing data corruption under heavy load.
- Transitioned bare-metal super-loop systems to an RTOS-based architecture, improving modularity and increasing CPU idle time by an estimated 95%.
- Led hardware bring-up and debugging across 30+ custom PCB iterations in coordination with the hardware team, cutting development cycles by 60%.
- Architected the application structure and custom build system for a residential Data Concentrator Unit, deploying custom Embedded Linux images that manage edge data collection for 120+ smart grid nodes.
- Built system initialization and RTC synchronization services, reducing boot time from a 4-minute legacy vendor average to 45 seconds.

Embedded Systems Part time Instructor

Smart Technology

Jun 2025 – Present

- Authored 20+ practical challenges for the ATmega32 MCU, covering GPIO, Timers, PWM, and External Interrupts.
- Delivered instruction and C/C++ code reviews to nearly 60 students, strengthening their bare-metal architecture understanding.

Professional Development:

Machine learning for Embedded Systems

Udemy

June 2026

- Deployed end-to-end TinyML applications on resource-constrained microcontrollers, covering the full pipeline from model training to on-device inference.
- Applied hardware-aware model optimization techniques for Edge AI, targeting Cortex-M class processors and dedicated neural processing units.
- Applied model compression techniques, including quantization, weight pruning, and clustering, to run memory-bound deep learning models within tight embedded hardware constraints.

The Complete Prompt Engineering for AI Bootcamp (2026)

Udemy

April 2026

- Applied prompt design techniques, including zero-shot, few-shot, and chain-of-thought prompting, to improve LLM performance and accuracy.
- Developed strategies for applying generative AI tools to real-world problem-solving through iterative prompt tuning and role-based prompting.
- Applied AI ethics and risk-mitigation practices to reduce model bias and hallucinations.

Embedded Linux

EDGES

September 2024

- Built Embedded Linux systems using Yocto, Buildroot, and manual build processes.
- Developed and deployed Linux applications in C++ using OOP principles, with custom Makefile and CMake build systems.
- Wrote Bash scripts for automation, system setup, and debugging.
- Built custom embedded Linux environments, integrating drivers, services, and user-space components..

Education:

Bachelor Degree in Systems and Biomedical Engineering

El Shorouk Academy, Cairo, Egypt

2015 – 2020

Core Skills:

Programming Languages: C / Embedded C (primary), C++ (OOP, Linux applications), Python, Bash.

Communication Protocols: SPI, I2C, UART, CAN, LIN, PLC (Power Line Communication), Ethernet, Wireless (BLE, WI-FI, Zigbee).

Embedded Systems: RTOS, Embedded Linux, Bare-metal, Device Drivers, Bootloaders, ARM Cortex-M, Firmware Development, Memory Management, Hardware Bring-up.

Tools: Yocto, Buildroot, GDB, JTAG, SWD, Logic Analyzer, Oscilloscope, Git, Makefile, CMake, CI/CD.

Language Skills:

- Arabic: Native speaker.
- English: Advanced level.